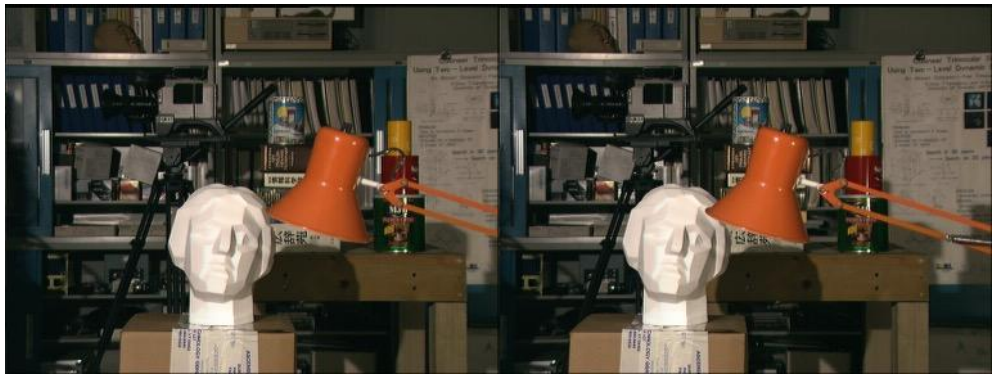
CS5187: Vision and Image

Multi-view stereo

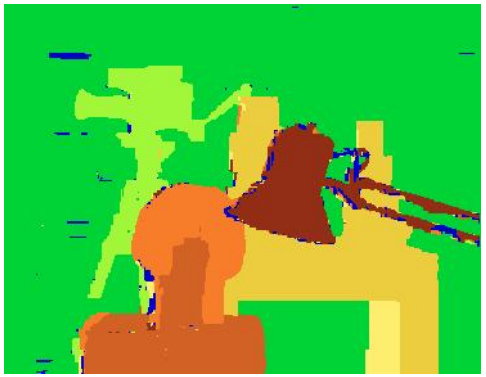


Multi-Camera Array

Before: Binocular (Two-View) Stereo



Left-right (rectified) stereo pair



Computed disparity map

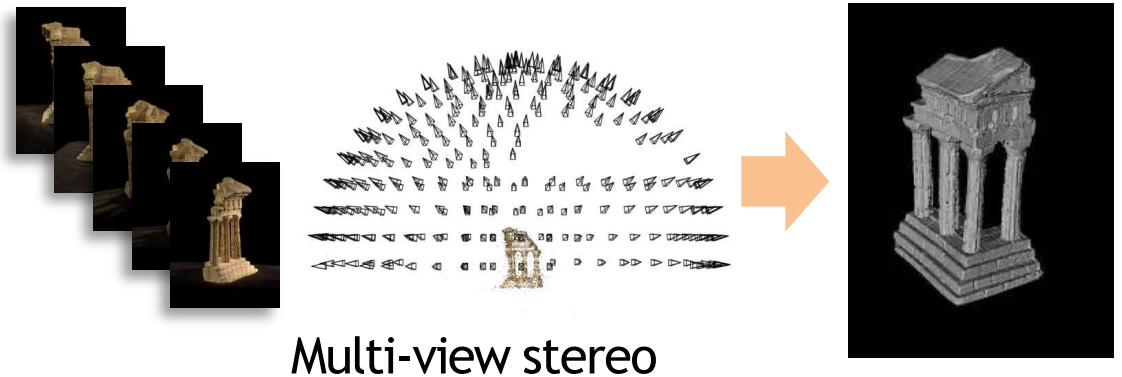
Useful for robot perception and navigation, video effects, etc.

Now: Multi-view Stereo

Problem formulation: given several images of the same object or scene, compute a representation of its 3D shape



Binocular Stereo



Multi-view stereo

Multi-view Stereo



Point Grey's Bumblebee XB3



Point Grey's ProFusion 25

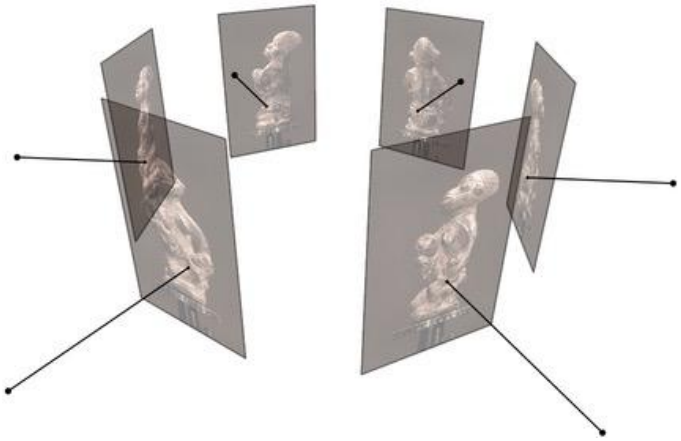


CMU's Panoptic Studio

Multi-view Stereo

Input: calibrated images from several viewpoints (known intrinsics and extrinsics / projection matrices)

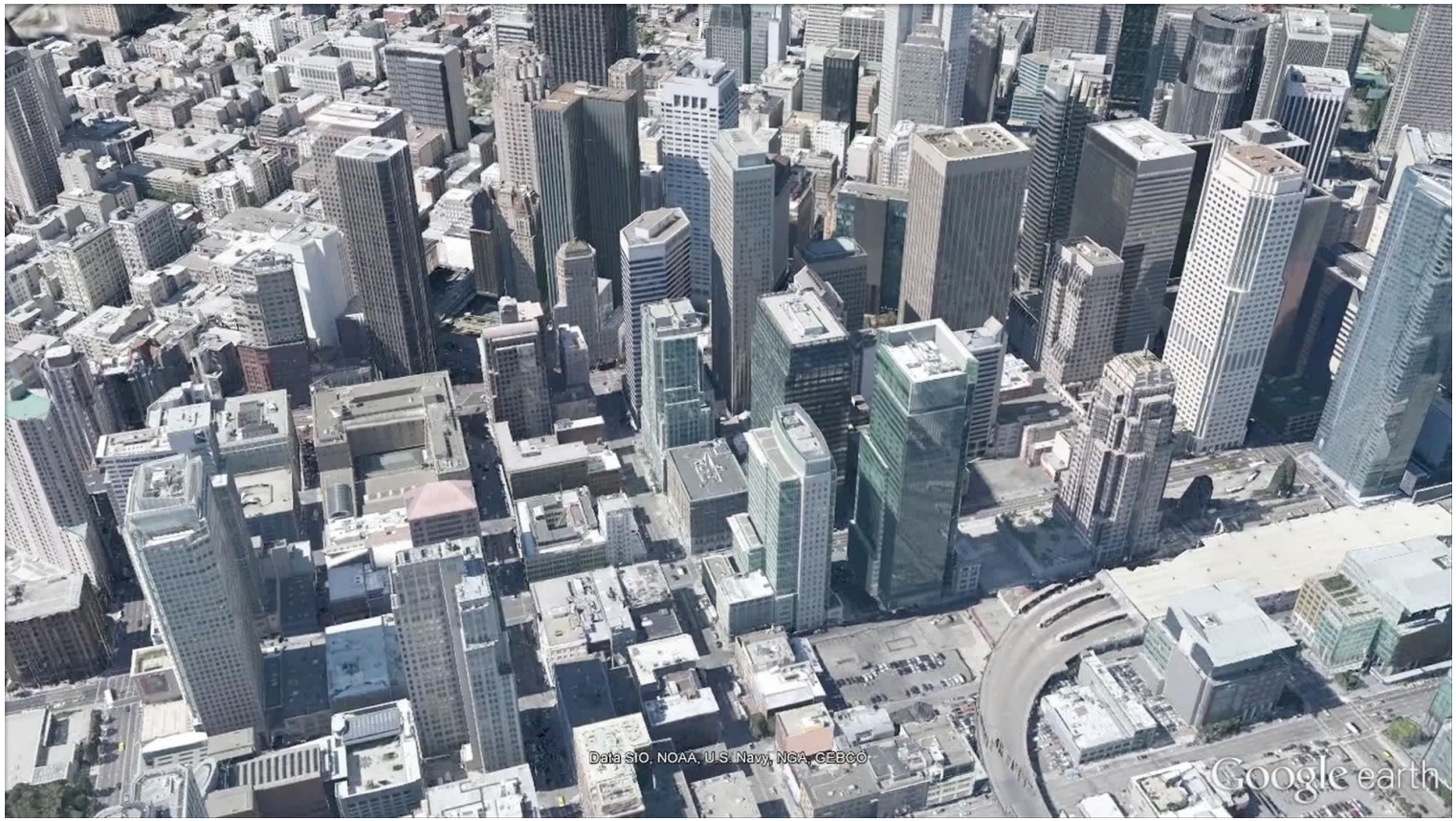
Output: 3D object model



Figures by Carlos Hernandez

We'll talk more about how to calibrate multiple cameras soon

Applications


Data SIO, NOAA, U.S. Navy, NGA, GEBCO
Google earth

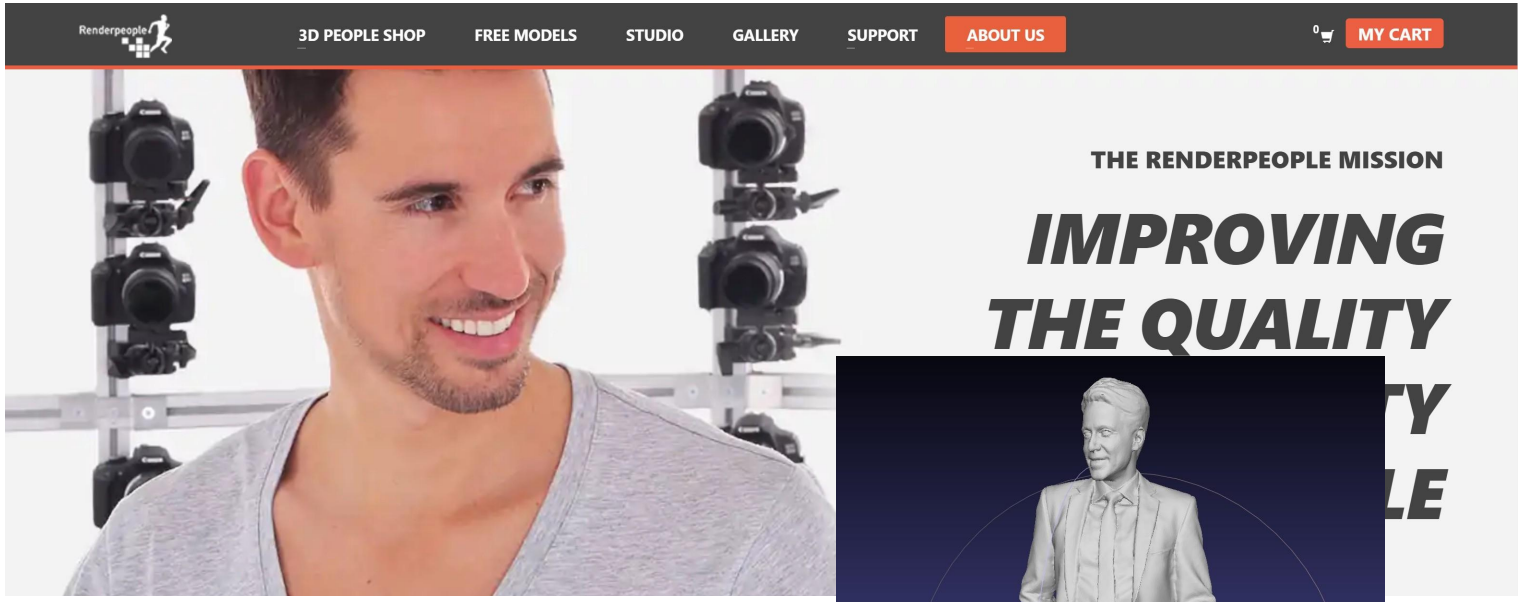
Whistle in the Form of Female Figure *600 AD - 900 AD*

Details Los Angeles County Museum of Art



Los Angeles County Museum of Art | Sculpture | Mexico

Share | Compare | Saved | Discover | Google



https://renderpeople.com/about-us/



Application: Virtual Reality Video



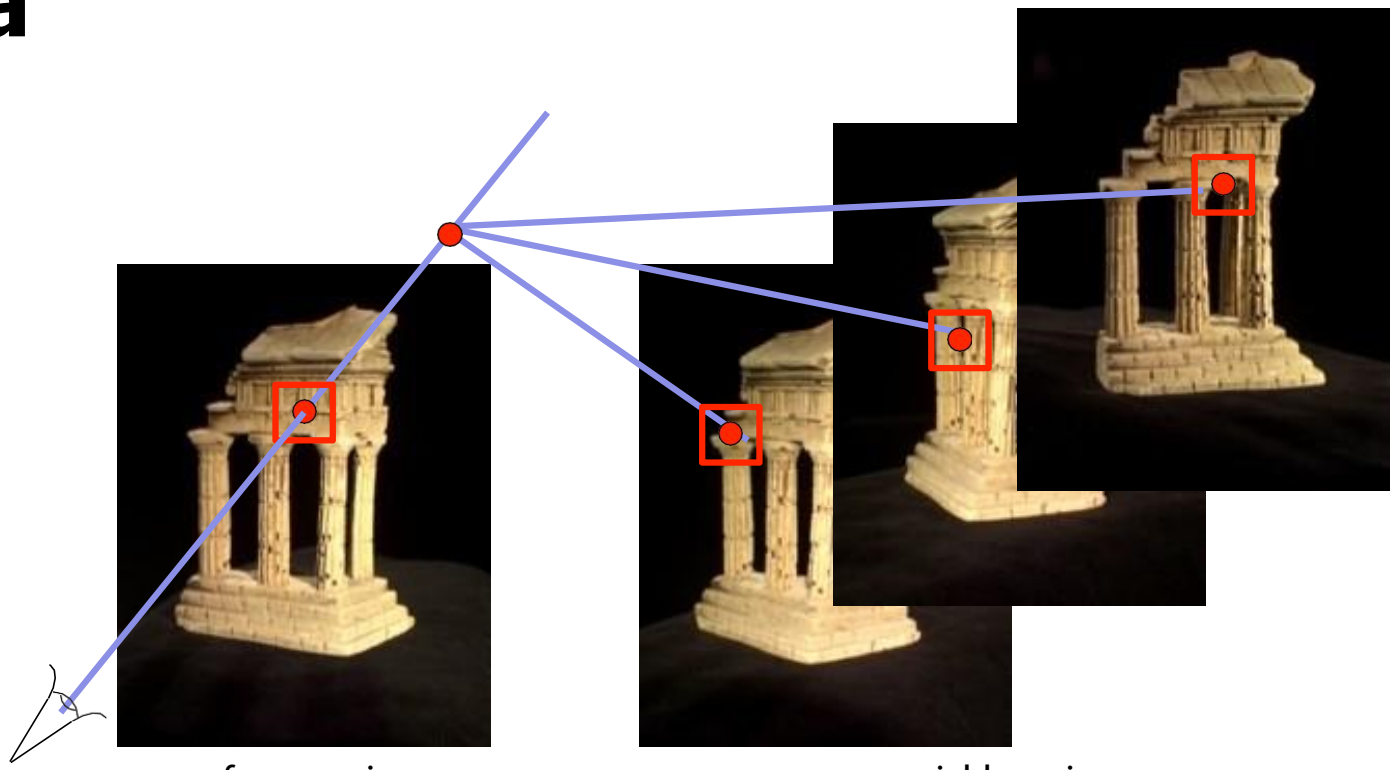
Anderson, et al. *Jump: Virtual Reality Video*. SIGGRAPH Asia 2016.



Broxton, et al. *Immersive Light Field Video with Layered Mesh Representation*. SIGGRAPH 2020



Multi-view stereo: Basic idea

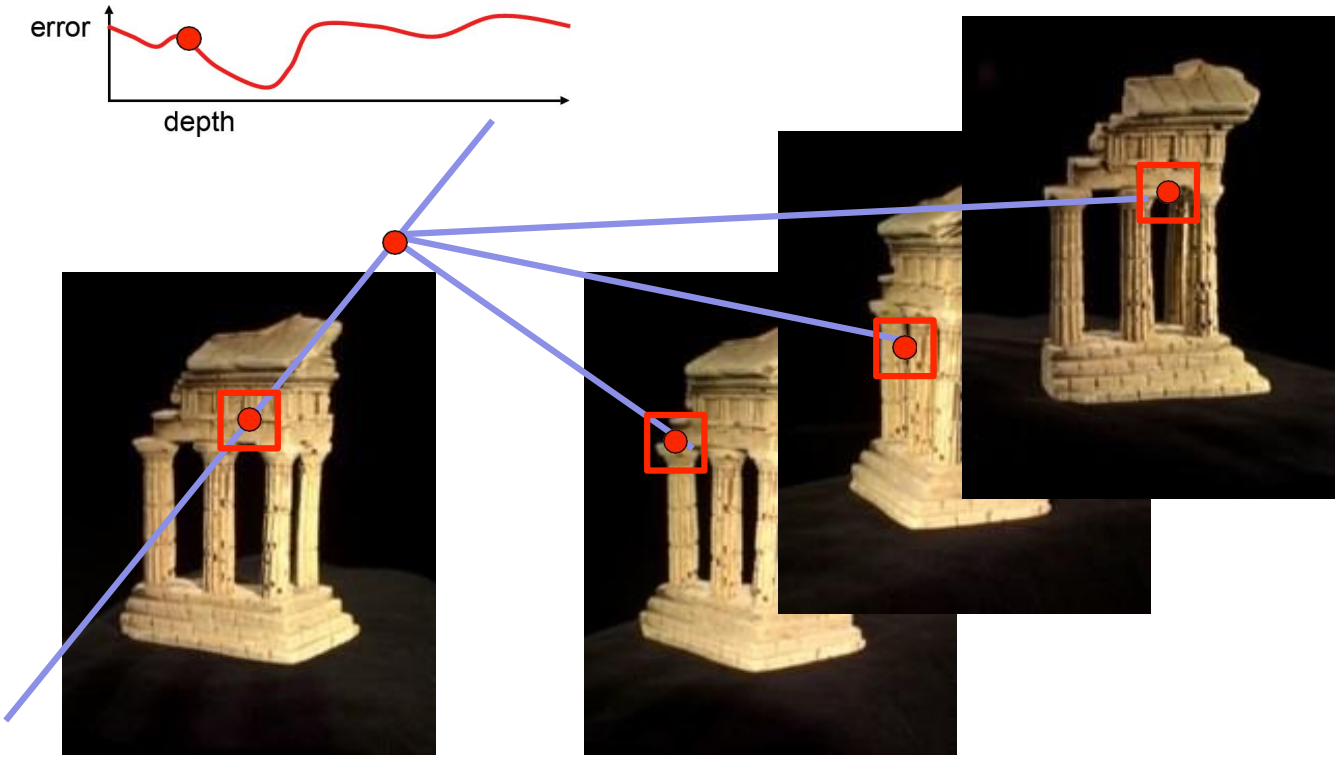


reference view

neighbor views

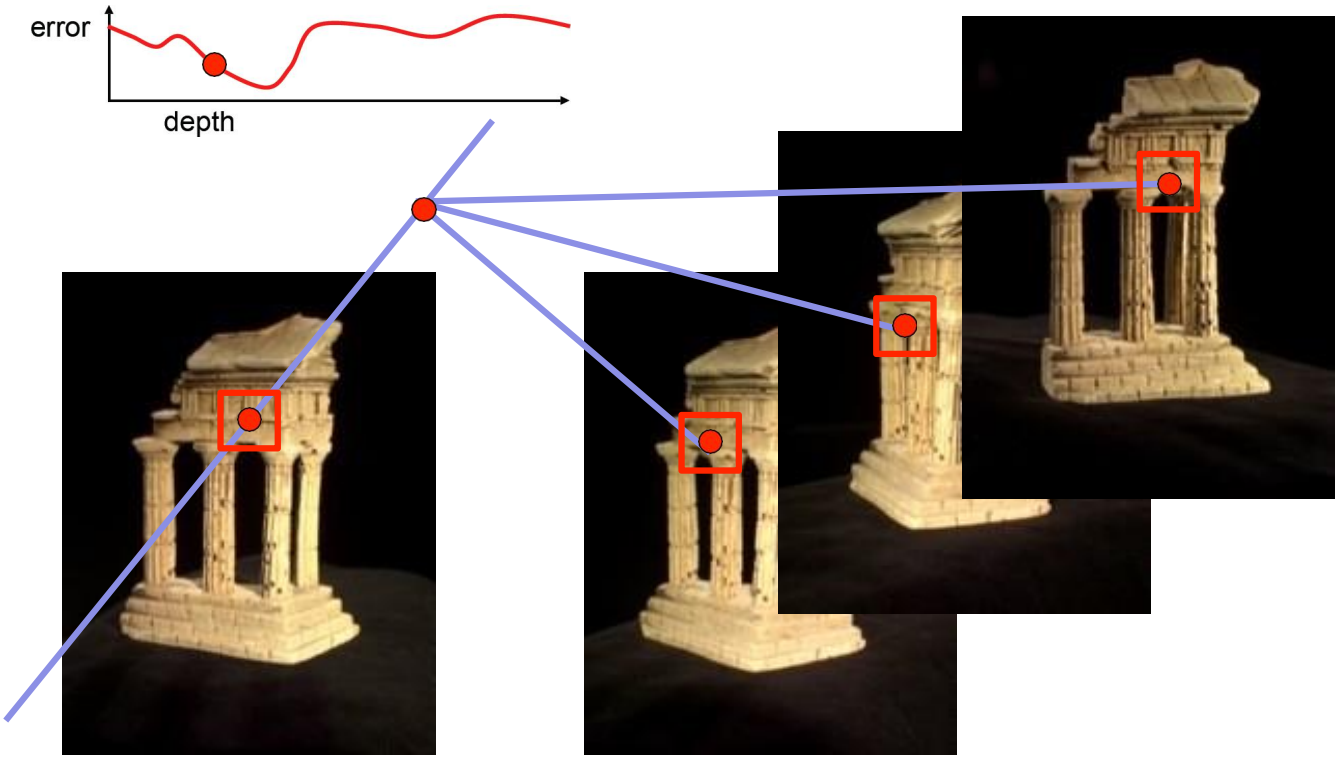
Source: Y.Furukawa

Multi-view stereo: Basic idea



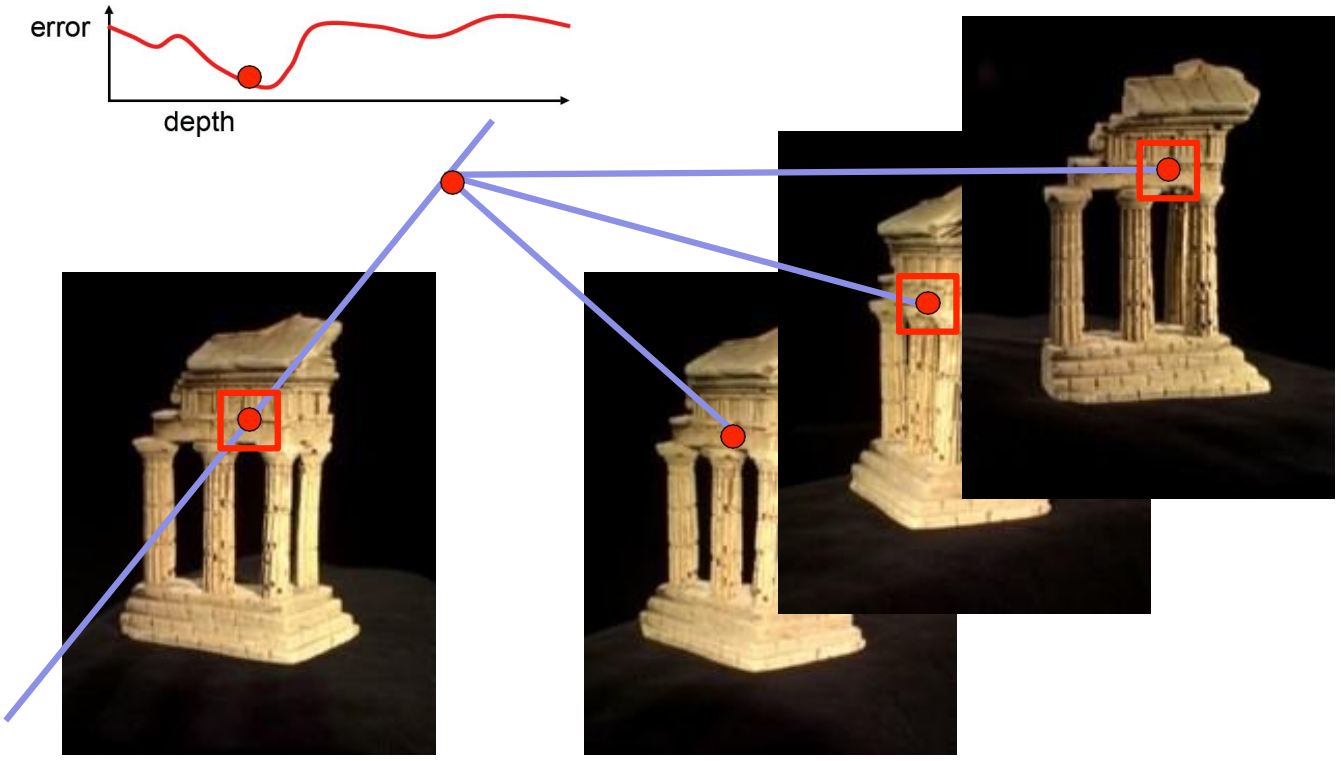
Source: Y.Furukawa

Multi-view stereo: Basic idea



Source: Y.Furukawa

Multi-view stereo: Basic idea



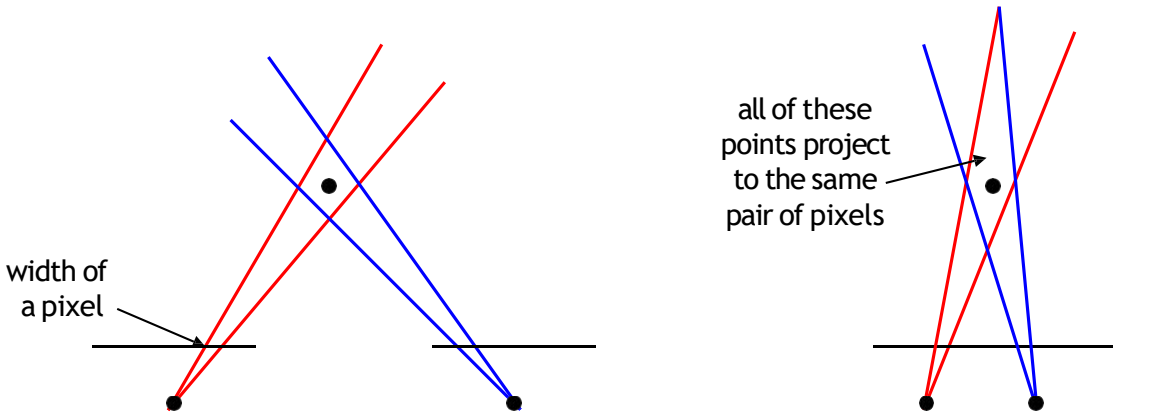
In this manner, solve for a depth map over the whole reference view

Source: Y.Furukawa

Multi-view stereo: advantages

- Can match windows using more than 1 neighbor, giving a **stronger match signal**
- If you have lots of potential neighbors, can **choose the best subset** of neighbors to match per reference image
- Can reconstruct a depth map for each reference frame, and the merge into a **complete 3D model**

Choosing a single stereo baseline



Large Baseline **Small Baseline**

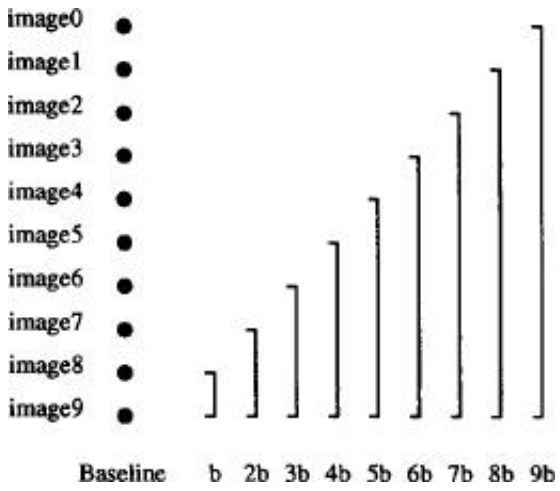
What's the optimal baseline?

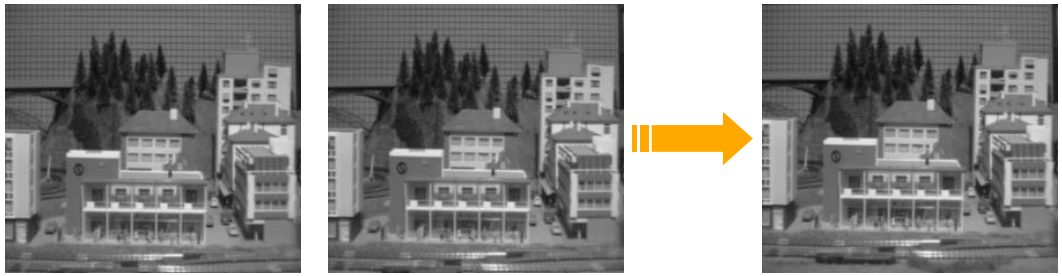
- Too small: large depth error
- Too large: difficult search problem

The Effect of Baseline on Depth Estimation



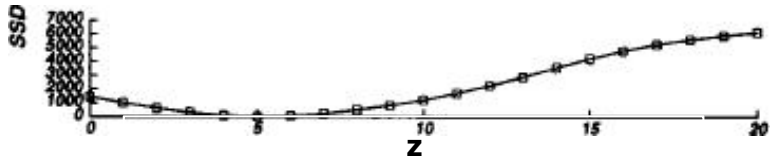
Figure 2: An example scene. The grid pattern in the background has ambiguity of matching.



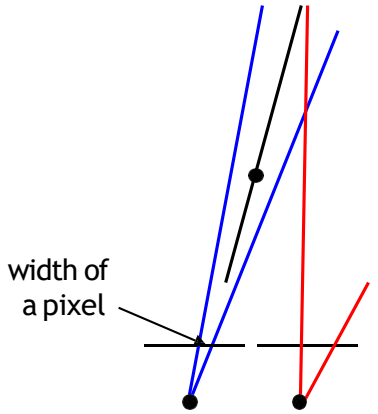


I_1 I_2 I_{10}

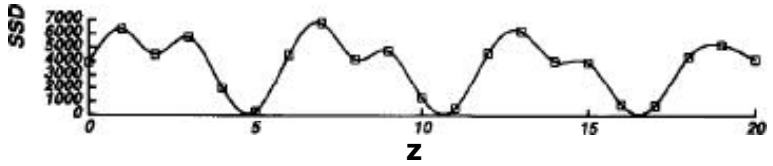
Multiple-baseline stereo – all cameras in a row



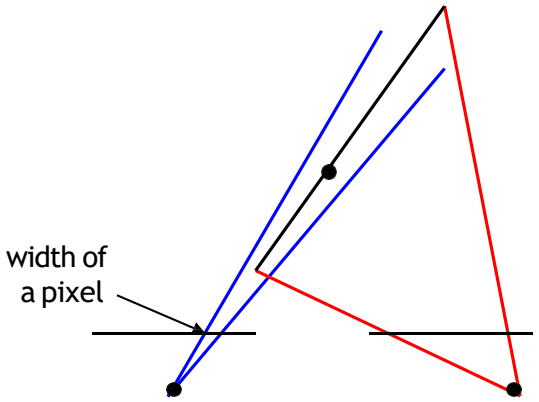
pixel matching score



- For short baselines, estimated depth will be less precise due to narrow triangulation



- For larger baselines, must search larger area in second image



M. Okutomi and T.Kanade, "A Multiple-Baseline Stereo System," IEEE Trans. on Pattern Analysis and Machine Intelligence, 15(4):353-363 (1993).

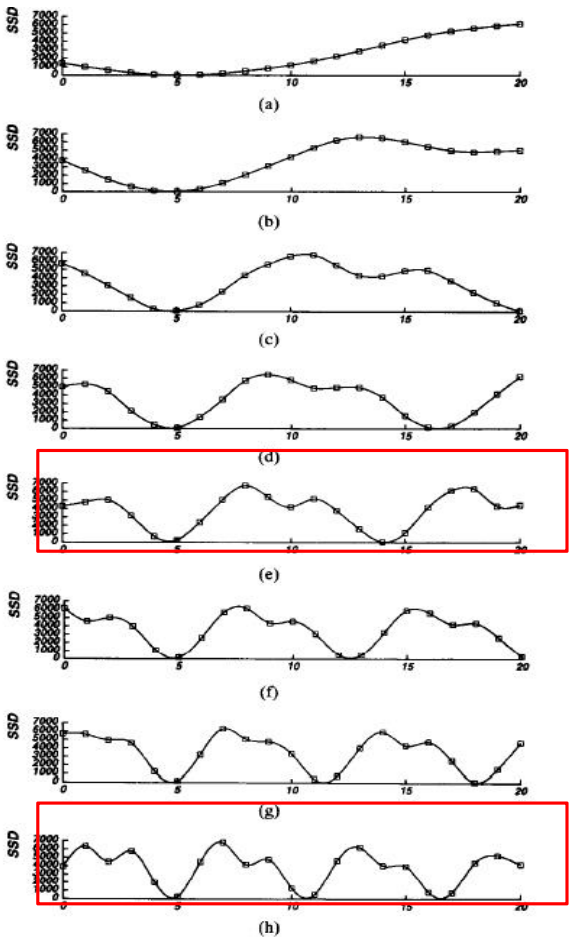


Fig. 5. SSD values versus inverse distance: (a) $B = b$; (b) $B = 2b$; (c) $B = 3b$; (d) $B = 4b$; (e) $B = 5b$; (f) $B = 6b$; (g) $B = 7b$; (h) $B = 8b$. The horizontal axis is normalized such that $8bF = 1$.

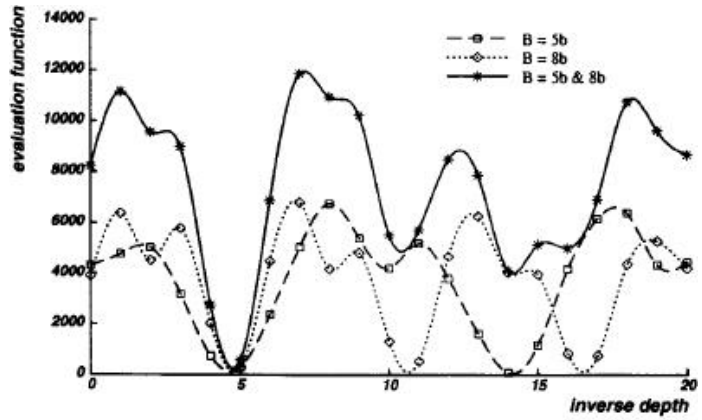


Fig. 6. Combining two stereo pairs with different baselines.

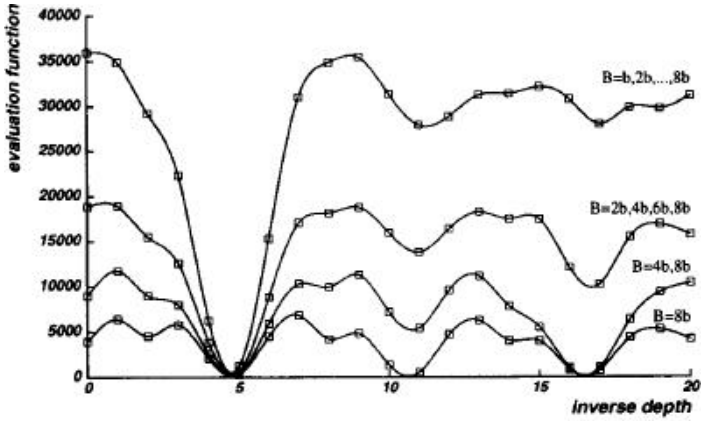
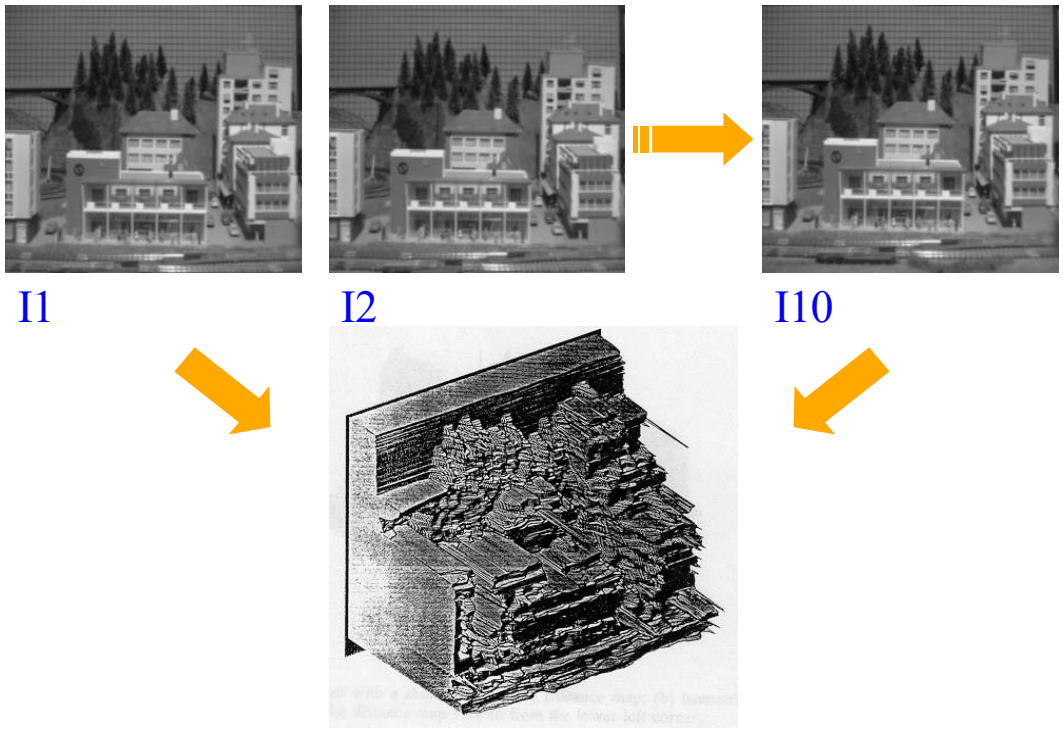


Fig. 7. Combining multiple baseline stereo pairs.

Multiple-baseline stereo results



M. Okutomi and T.Kanade, *A Multiple-Baseline Stereo System,* IEEE Trans. on Pattern Analysis and Machine Intelligence, 15(4):353-363 (1993).

Multibaseline Stereo

Basic Approach

- Choose a reference view
- Use your favorite stereo algorithm BUT
  - replace two-view SSD with **SSSD** over all baselines
  - **SSSD**: the SSD values are computed first for each pair of stereo images, and then add all together from multiple stereo pairs.

Limitations

- Only gives a depth map (not an "object model")
- Won't work for widely distributed views.

Problem: *visibility*

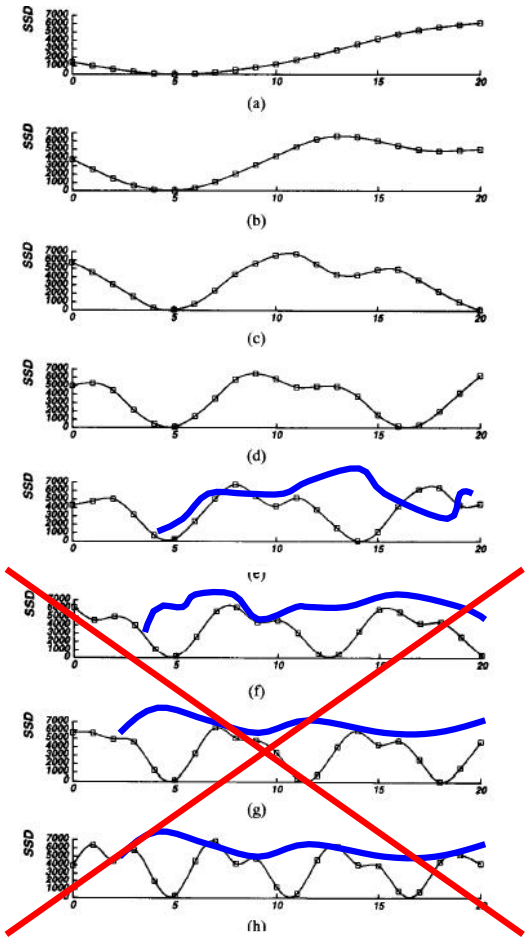


Fig. 5. SSD values versus inverse distance: (a) $B = b$; (b) $B = 2b$; (c) $B = 3b$; (d) $B = 4b$; (e) $B = 5b$; (f) $B = 6b$; (g) $B = 7b$; (h) $B = 8b$. The horizontal axis is normalized such that $8bF = 1$.

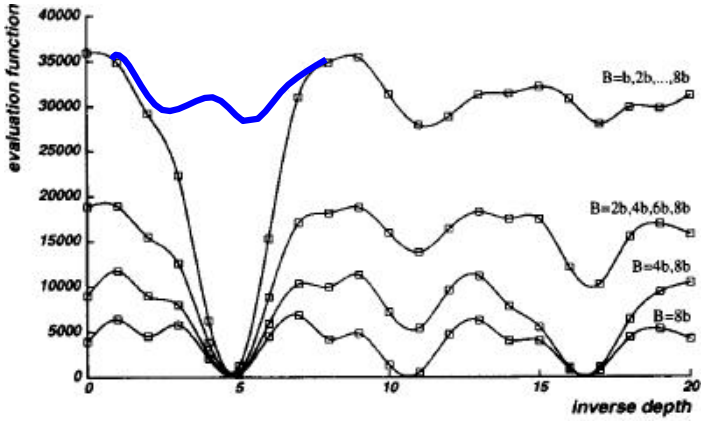


Fig. 7. Combining multiple baseline stereo pairs.

Some Solutions

- Match only nearby photos [Narayanan 98]
- Use NCC instead of SSD,
  Ignore NCC values > threshold [Hernandez & Schmitt 03]

Popular matching scores

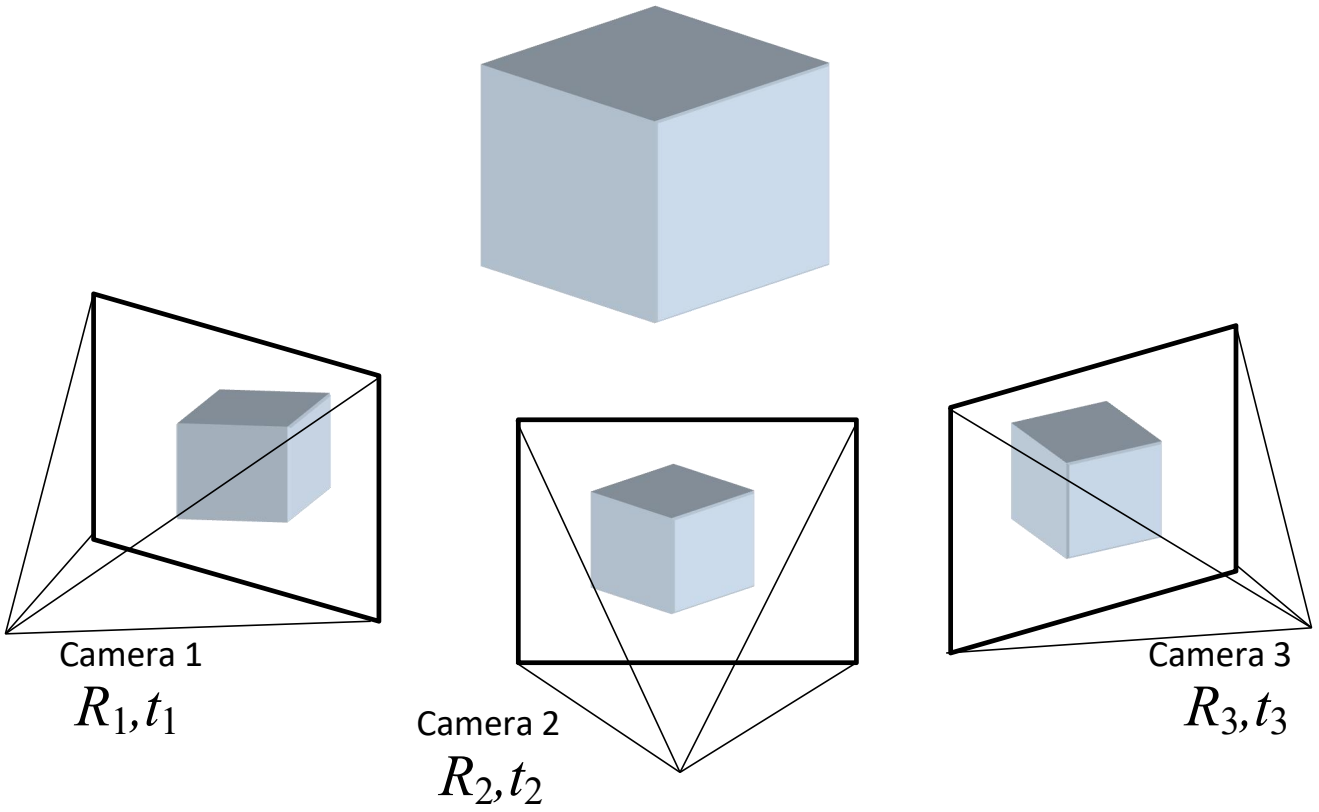
- SSD (Sum of Squared Differences) $\sum_{x,y} |W_1(x,y) - W_2(x,y)|^2$
- SAD (Sum of Absolute Differences) $\sum_{x,y} |W_1(x,y) - W_2(x,y)|$
- ZNCC (Zero-mean Normalized Cross Correlation)

$$\frac{\sum_{x,y}(W_1(x,y) - \overline{W_1})(W_2(x,y) - \overline{W_2})}{\sigma_{W_1}\sigma_{W_2}}$$

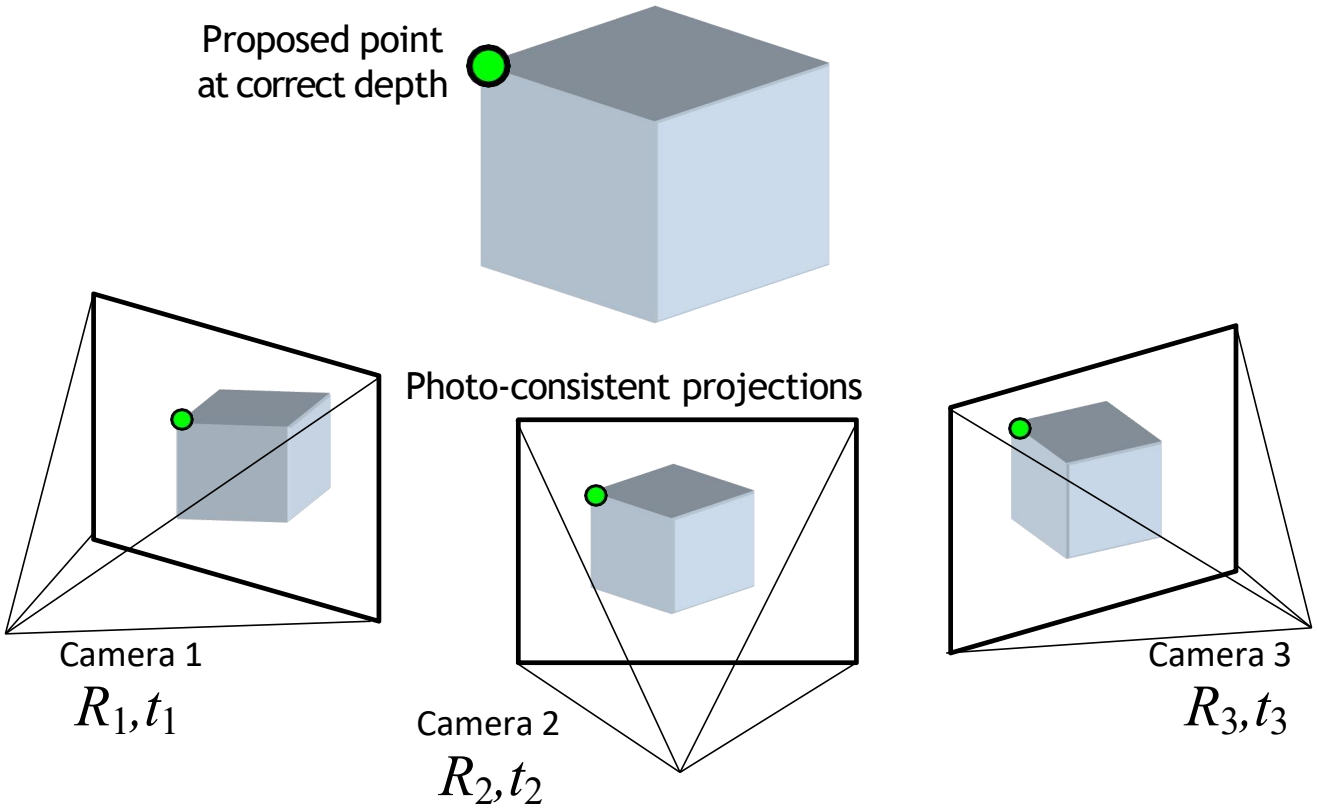
  - where $\overline{W_i} = \frac{1}{n}\sum_{x,y} W_i \qquad \sigma_{W_i} = \sqrt{\frac{1}{n}\sum_{x,y}(W_i - \overline{W_i})^2}$
  - what advantages might NCC have?

Questions?

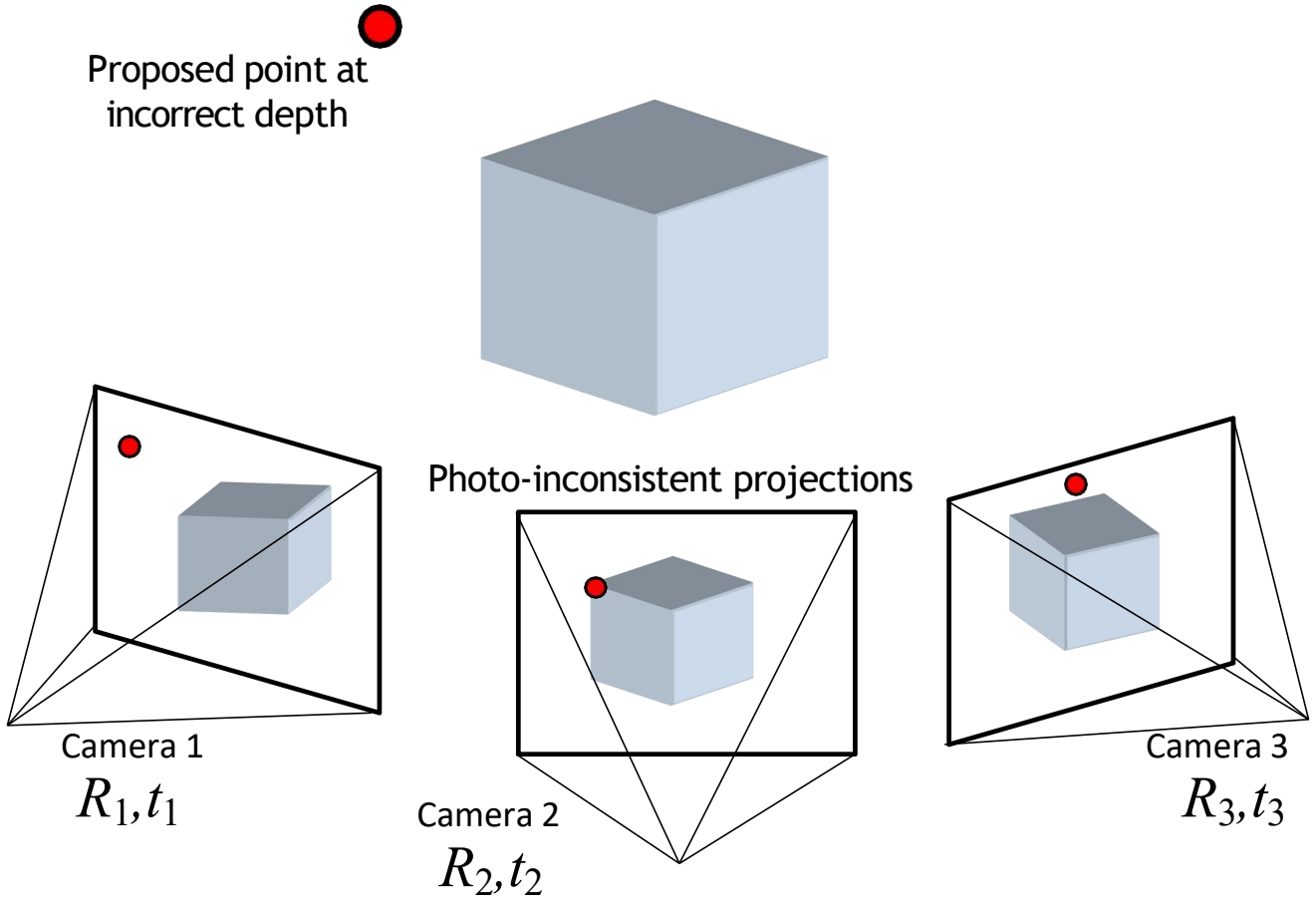
Plane-Sweep Stereo



Plane-Sweep Stereo

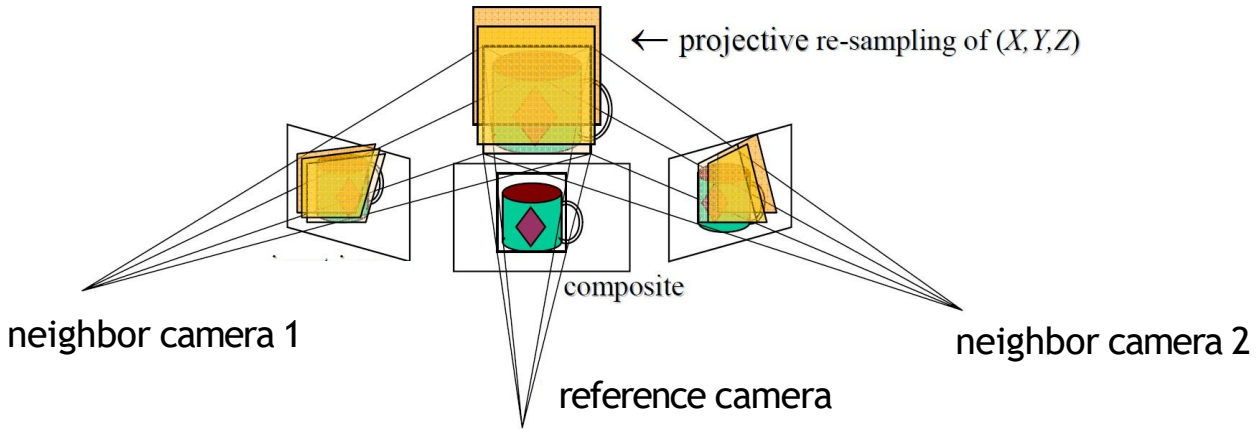


Plane-Sweep Stereo



Plane-Sweep Stereo

- Sweep family of planes parallel to the reference camera image plane
- Reproject neighbors onto each plane (via homography) and compare reprojections



Plane-Sweep Stereo



Left neighbor



Reference image



Right neighbor



Left neighbor projected into reference image

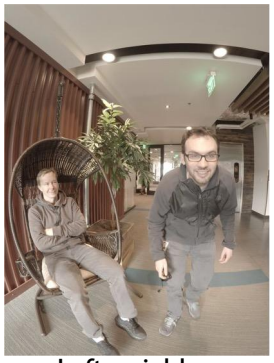


Average images on each plane



Right neighbor projected into reference image

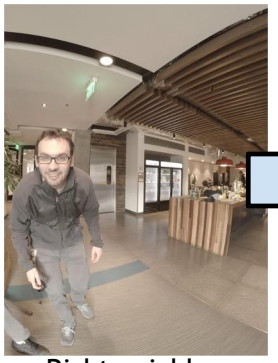
Another example



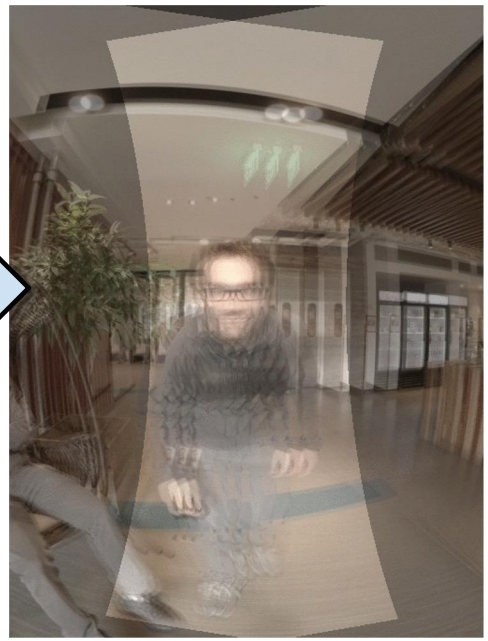
Left neighbor



Reference image

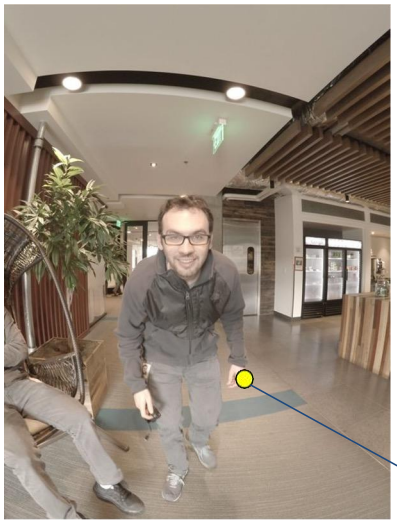


Right neighbor



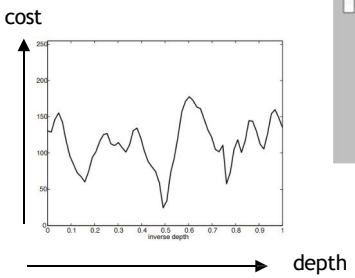
Planar image reprojections swept over depth (averaged)

Cost Volumes -> Depth Maps

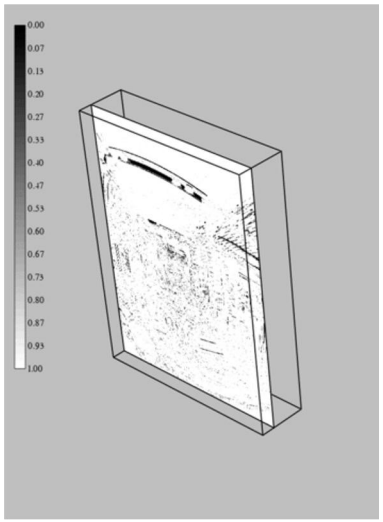


Reference image

Plane sweep



Single pixel's cost profile



Full cost volume

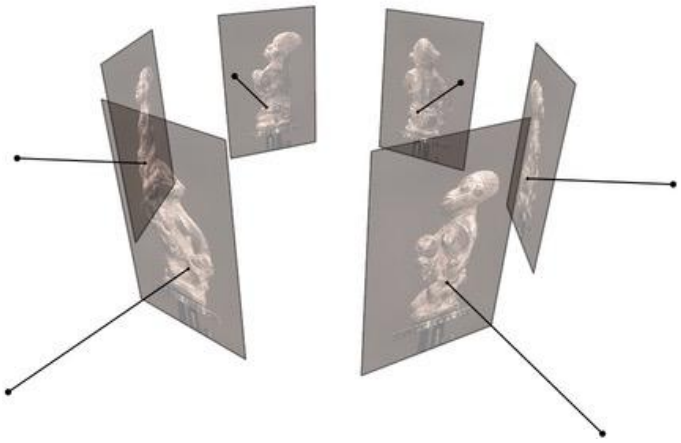
Depth map solver

(Belief propagation, graph cuts, etc.)



Fusing multiple depth maps

- Compute depth map per image
- Fuse the depth maps into a 3D model



Figures by Carlos Hernandez

Another approach: NeRF

- Represent scenes as functions from (x, y, z) to RGB and alpha (transparency), use volume rendering to render images





NeRF: Representing Scenes as Neural Radiance Fields for View Synthesis, ECCV 2020

https://www.matthewtancik.com/nerf

Questions?